

Mentee ID: #01 - 250101

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TEXT PREPROCESSING ASSIGNMENT REPORT

Vietnamese Online News Text Preprocessing

Preprocessing Steps

Before applying the main pipeline, I first **removed empty news content** from the raw dataset. Then, I implemented a **Processor class** to perform the following **end-to-end text processing steps**:

- **Text normalization**: convert all characters to lowercase.
- **RegEx cleaning**: remove unnecessary elements, including:
 - o Email addresses
 - o URLs and domains (.com, .net, .vn, ...)
 - o Copyright notes
 - o Numbers
 - o Punctuation
 - o Extra white spaces
- **Tokenization**: apply **underthesea** for accurate Vietnamese word segmentation, this is the NLP toolkit in Python for NLP like NLTK, but in Vietnamese language.
- **Stopword removal**: use Vietnamese stopwords list (widely used dataset in community from Kaggle/GitHub).
- **Output**: save processed text into a new file for later usage.

(Because the nature of Vietnamese is different from English, I eliminated the lemmatization step from the pipeline)

Additionally, in the **full pipeline**, an `early_stopping` parameter was added to limit the number of processed documents for fast notebook experiments.

Results

I tested for the first 1000 documents in the dataset and observed the statistical results (*performed in experiment notebook*) after processing as follows:

- **Cleaner text** with consistent normalization and element removal.
- Significant reduction in text length led to smaller vocabulary size.
- Text more **optimized for vectorization/embedding** steps.

- **Underthesea** ensures meaningful Vietnamese tokenization:
 - Produces consistent tokens (e.g., "công an" to công_an).
 - Effectively reduces noise and ambiguity.

Key Insights & Reflections

- Regex rules must be carefully designed; too strict patterns may remove useful tokens.
- Stopword lists need domain adaptation (general list might remove some important words in news context). Further development in create own stopwords list for text preprocessing case in news.
- underthesea works well but still has limitations on slang and abbreviations, need possible manual corrections.
- Early stopping is useful for fast prototyping before scaling to the entire dataset.